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HCI- 5840
Fall 2026

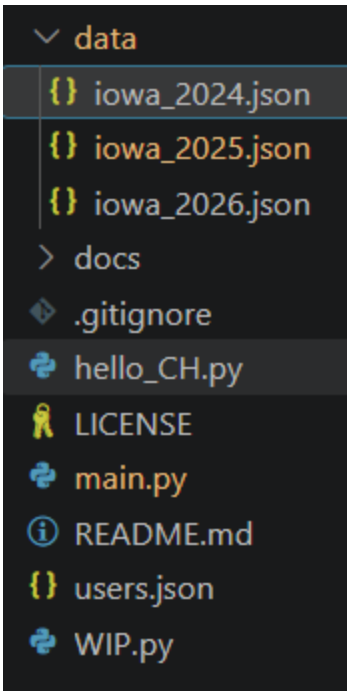
Safe Driving Buddy – Version 1 Review

Project overview:
Safe Driving Buddy is a project that helps drivers learn about traffic crash information before traveling to an unfamiliar location within Iowa. The goal is to retrieve crash data from the NHTSA FARS API and present useful safety information to users. Eventually, the project may include a web interface and map visualization, but the current version focuses on retrieving and processing crash data through Python.
So far, I have implemented the following:

- Retrieved crash data from the NHTSA FARS API.
- Saved API responses as JSON files.
- Created a Python program that reads JSON data.
- Displayed crash information such as the API message and crash count in the terminal.
- Organized project files using a data folder and documentation folder.

Screenshots of the app currently working:

Crash data is downloaded from the NHTSA FARS API and saved as JSON files.



The Python program opens a JSON file.

```python
import json
# Open and read the JSON file containing Iowa 2024 data
with open("data/iowa_2024.json", "r", encoding="utf-8") as file:
    data = json.load(file) # Convert the JSON file content into a Python dictionary

print("Message:", data["Message"]) # Display the message return by the API
print("Count:", data["Count"]) # Display the total number of crash records
```

The program reads the data and displays information in the terminal.

```
PS C:\Users\sacharya\Documents\GitHub\Samjhana_DrivingBuddy> python main.py
Message: Results returned successfully
Count: 2898
```

Issue encountered:
At the beginning of the project, I didn't have much idea about API data or how it works. By repeating the task, I am now a little more comfortable requesting, downloading, and storing JSON data. The API currently has no crash data for 2025 or 2026 and only has data for 2024.

That is why I tried using the older data from 2022 and 2023, but the API started returning a 403 Forbidden error. This issue is not yet resolved. I will continue trying to download the data.

Unsolved Issue:

The API is currently returning a 403 Forbidden error when I try to access some requests. My plan is to investigate the issue and continue developing and working with 2024 data.

Help:

I would appreciate your help resolving the 403 Forbidden API error. Since this is my first time working with API data, I am not sure whether this error is normal. Also, I'd appreciate tips on the best way to organize crash data for analysis.

Milestones for version 2:

1. Improve crash data processing. I will use Python to read and organize crash data from the JSON files. Share useful information and prepare the data for analysis.
2. Generate Crash Summaries. Create summary statistics from the crash data, such as the total number of crashes and other available information from the NHTSA data. Display these summaries in a clear and easy-to-read format.
3. User Selection Feature. Allow users to select a year and view available crash information from the corresponding data file.
4. Improve the user Interface. I will create a simple command-line menu that lets users select available crash data and view results without directly modifying the Python code.

Self-reflection:

So far, I feel fine about the progress. I have completed the core requirement of retrieving crash data and reading it with Python. I learned how to use GitHub, which was a big task for me.

I think I can finish the project in the remaining weeks. From now on, I need to focus on processing and presenting the data.

My expectation vs. reality was that working on the project from scratch can feel a little overwhelming at times, especially when working with API data. The reality is that once you break the steps down, it becomes easier to accomplish.

My light bulb moment is that the more you do, the more comfortable and confident you feel.